FIRE-MODEL Proposal Draft v3

NSF-Aligned Full-Length Narrative Draft

Working assumptions. This draft is structured to satisfy NSF narrative expectations while aligning with the FIRE-MODEL track framing and the internal strategy documents in this repository. Budget and budget justification are intentionally excluded because those are prepared through separate submission forms. The text below is intentionally detailed and near full length so that revision cycles can focus on precision and evidence support rather than discovering missing structure late.

Project Summary (Target: 750 words total)

1) Overview

Target length: 300 words. Wildfire behavior is increasingly shaped by interacting extremes: prolonged fuel aridity, compound meteorological forcing, and rapid shifts in atmospheric transport conditions. Although current operational and research models provide useful guidance in many scenarios, they often become least reliable in the regimes where consequences are most severe. This mismatch arises in part because many existing approaches are calibrated around local spread behavior or static assumptions, while large events can evolve into dynamically distinct stages with different dominant controls.

This proposal addresses that gap by developing a transition-aware, multi-regime wildfire modeling framework grounded in measurable quantities. Rather than assuming one universal growth law across all fires and all times, we test the hypothesis that wildfire progression moves through identifiable regimes: pre-emergent spread, transition to coherent front organization, and post-emergent boundary-dominated growth. We combine event-based observations from `FIRED`, scalable analysis infrastructure from `CubeDynamics`, and a reproducible measurement-model-validation workflow to determine when each regime appears, what mechanisms are sufficient in each regime, and how uncertainty changes across regime boundaries.

The project's core products are empirical and computational: a curated event-level dataset with geometry and coupling metrics; open-source scaling and model-comparison diagnostics; a transition-aware benchmark suite spanning observations and simulations; and validated uncertainty reporting templates for decision-relevant interpretation. The intended outcome is not a single "winner" model, but a framework that narrows plausible mechanisms, clarifies model limits, and provides a transparent path from physical hypothesis to robust comparative evidence.

Practically, this project is designed as a discovery program with explicit decision points. We specify what evidence would support or challenge each claim, what metrics must be stable across datasets before we generalize, and how negative findings are handled. This creates a scientifically honest workflow in which uncertainty and failure modes are treated as knowledge, not obstacles.

2) Intellectual Merit

Target length: 225 words. The intellectual merit lies in unifying wildfire geometry, process dynamics, and uncertainty quantification within a falsifiable inferential framework. The project contributes three linked advances. First, it formalizes measurable regime-transition diagnostics, including continuity, geometry, and transport-demand proxies, allowing explicit tests of when local spread assumptions cease to be sufficient. Second, it develops a hybrid modeling architecture that combines local process realism with reduced geometric representations in regime-dependent ways. Third, it establishes a strict verification and validation pipeline that evaluates not only predictive error, but structural fidelity, identifiability, and calibration under changing forcing.

Methodologically, the proposal integrates event-based geospatial analytics, stochastic and mechanistic model experiments, and multi-objective comparison metrics including distributional and time-series diagnostics. It extends beyond pointwise forecasting by asking whether models reproduce observed cross-scale structure, such as perimeter-area scaling behavior, interface roughening, and regime-specific dynamics. By explicitly defining failure criteria and reporting negative outcomes transparently, the project advances wildfire science as a mechanism-elimination program rather than a tuning exercise.

The deeper intellectual contribution is a shift in how wildfire models are judged. Instead of rewarding aggregate fit alone, we test whether models explain the observed organization of fire growth across scales and conditions. This reframing supports theory building, not just prediction contests.

3) Broader Impacts

Target length: 225 words. Broader impacts center on translation, workforce development, and open scientific infrastructure. The project will release reproducible tools, benchmark datasets, and method documentation so that external groups can evaluate model behavior with a shared analytical language. This reduces duplication, improves comparability across studies, and lowers entry barriers for institutions that lack large internal modeling teams.

Training is embedded in project execution through structured roles for postdoctoral researchers and interdisciplinary collaboration across geospatial data science, complex systems modeling, and uncertainty analysis. Trainees will build publishable systems, lead conference outputs, and gain experience in transparent workflow design, strengthening the wildfire modeling workforce.

The project also includes public-facing translation pathways described in repository planning documents: interactive exploratory tools, educational materials, and narrative media partnerships designed to communicate mechanism-based wildfire understanding to broader audiences. The broader-impact goal is to move beyond one-way dissemination toward reusable, feedback-informed infrastructure that supports scientists, agency research groups, and long-term downstream decision systems.

We will evaluate impact through adoption and use, not only dissemination counts: external reuse of code and datasets, workshop participation outcomes, educational integration, and evidence that uncertainty-aware outputs change how wildfire modeling results are interpreted.

Project Description (Target: 9,000 words total)

1. Introduction, Motivation, and Opportunity

Target length: 800 words. Wildfire science now operates in a paradox. On one hand, data richness has expanded dramatically through satellite burned-area records, event reconstruction systems, and environmental reanalysis products. On the other hand, explanatory and predictive confidence remains fragile in the exact conditions where stakes are highest: rapid growth windows, compound forcing, and large coherent fronts. Existing model classes are often evaluated against local rates or final extent, yet those endpoints can conceal whether the model captured the process transitions that generated the event trajectory.

This proposal treats that gap as a scientific opportunity. Repository strategy documents emphasize a shift from "single-law" framing to a regime-based framing in which wildfire can transition between states with different effective controls. In pre-emergent phases, spread may be patchy, ignition-limited, and strongly local. During transition, continuity and acceleration can change rapidly as coupling strengthens. In post-emergent phases, interface organization and geometry may become dominant constraints. A valid model class should therefore be judged by regime-specific adequacy, not by aggregate fit alone.

The FIRE-MODEL track calls for innovative methods and algorithms across disparate spatiotemporal scales, measurable pathways for model improvement, and holistic verification and validation with uncertainty quantification. Our design maps directly to these priorities. We build measurable transition diagnostics first, then test mechanism classes against those diagnostics, then package the workflow as reusable benchmark infrastructure. This sequence strengthens evidentiary clarity: what is observed, what is inferred, what each model must reproduce, and what constitutes rejection.

The project is also timely because foundational infrastructure already exists in this repository's scientific ecosystem: event reconstruction through `FIRED`, cube-based computational workflows through `CubeDynamics`, and an articulated methods ladder spanning measurement, model-data comparison, generative mechanism testing, and synthetic validation. The proposal does not begin from a blank slate; it leverages mature conceptual and computational scaffolding to pursue a decisive, falsifiable research program.

A second motivation is translational pressure. Agencies and communities increasingly encounter model outputs in planning and response contexts, yet uncertainty communication often remains underdeveloped. Overconfident interpretation is a practical risk. By distinguishing reducible from irreducible uncertainty and linking reliability to regime context, this project improves not only scientific understanding but the integrity of downstream interpretation.

A third motivation is cumulative science. Wildfire model development frequently proceeds with heterogeneous metrics, non-comparable preprocessing assumptions, and limited transparency around failure cases. The benchmark architecture proposed here is designed to make results portable and comparable across model families. This improves reproducibility and accelerates community learning.

Finally, this effort is positioned as a resubmission-strengthening program rather than a conceptual restart. The repository includes explicit reviewer-response framing: move from broad assertion

toward testable discriminators, define decisive benchmarks, and make success/failure criteria operational. This proposal is the implementation of that strategy.

A final motivation is interoperability across wildfire research communities. Different groups often maintain different preprocessing assumptions, variable names, and validation conventions, which makes synthesis difficult even when findings are individually strong. By building a common event-centered representation and shared benchmark language, this project helps align model evaluation practice across institutions.

The opportunity, therefore, is not merely to produce one improved model, but to establish a durable scientific protocol: observable-first hypothesis testing, model role clarity by regime, transparent uncertainty decomposition, and reproducible comparison artifacts that can be carried forward as new datasets and models emerge.

The project is explicitly theory-first generative modeling rather than simulation-first tuning. We begin from minimal governing principles that define wildfire growth as a process unfolding on a structured substrate, then test whether those principles are sufficient to reproduce observed fire forms. In this framing, fire form is treated as a measurable record of process: geometry, continuity, and scaling signatures are empirical constraints on mechanism validity, not decorative outputs.

2. Central Hypothesis and Specific Aims

Target length: 900 words. **Central hypothesis.** Large-fire growth is governed by transitions between local mechanism-dominated spread and coherent interface-dominated regimes; these transitions can be detected through measurable proxies and modeled with improved reliability using transition-aware hybrid frameworks.

Aim 1: Detect regime transitions and their controls. We will construct and evaluate observable indicators from event-level perimeter dynamics, geometry metrics, and atmosphere-fuel proxy families. A core test follows the ventilation-capacity versus combustion-demand framing from the resubmission strategy documents: whether regime transitions cluster near thresholds in proxy ratios such as V/D. Candidate indicators include continuity measures, change points in scaling exponents, roughness evolution, and temporal alignment between transition markers and transport-demand crossings.

Aim 2: Explain transitions through model contrasts. We will implement and compare model families with distinct assumptions: local spread and wave-like formulations, reduced geometric and manifold formulations for coherent fronts, and hybrid transition-aware mixtures. This aim evaluates which mechanisms are necessary in each regime and which simplifications are defensible without losing structural fidelity. The focus is not only forecast accuracy, but mechanistic sufficiency under explicit comparison protocols.

Aim 3: Apply results as a reusable benchmark and inference system. We will release an uncertainty-aware benchmarking pipeline that allows other researchers to score model outputs against shared observational diagnostics. Deliverables include versioned datasets, metric libraries, evaluation templates, and reproducible workflows suitable for extension across ecosystems and future model classes.

Falsifiable predictions. (1) Observable transition signals will show reproducible change points across multiple datasets and regions. (2) Transition-aware model architectures will outperform single-regime baselines in out-of-sample transition-heavy events. (3) Calibration-aware uncertainty reporting will improve reliability diagnostics and expose model-form limits that are obscured by aggregate error summaries.

Success and failure logic. Each aim includes explicit decision thresholds. If transition indicators lack cross-dataset robustness, inferential claims will be narrowed to validated domains. If hybrid models fail to improve calibrated performance or structural fidelity in targeted regimes, we will reject broad transition-aware superiority claims. This explicit failure logic is central to proposal credibility and aligns with the repository's commitment to transparent reporting of negative or inconclusive results.

Hypothesis testing structure. At the core is "Benchmark 0" from the strategy documents: compare single-law and multi-regime explanations under identical data and validation conditions. We will use AIC/BIC and related information criteria, cross-validation, explicit change-point comparison, and transition-alignment analysis testing whether inferred regime boundaries co-occur with independent transport-demand transitions such as crossings in V/D proxy families. This creates a decisive test rather than rhetorical contrast.

Regime classifier as product. Beyond narrative claims, we will deliver a versioned regime classifier that labels fire states as pre-emergent, transition, or post-emergent with uncertainty. This classifier is designed to be usable both for observed events and for simulated outputs, enabling direct model-observation comparability.

Alternative outcomes. The project is designed to remain informative under multiple outcomes. If evidence supports regime transitions but weakly supports specific proxy families, we refine transition detection while treating mechanism proxies as provisional. If no robust transitions are found, we report conditions under which single-regime assumptions remain adequate. In all cases, the benchmark outputs and reproducible pipelines remain valuable community assets.

Experiment matrix and crosswalk. Each aim maps to a concrete experiment matrix. For Aim 1, we test indicator robustness across event classes, ecosystems, and sensing conditions. For Aim 2, we evaluate model classes under matched forcing and boundary definitions to isolate mechanism differences. For Aim 3, we package these tests into reusable benchmark workflows with explicit metadata and reporting templates. This crosswalk ensures that every narrative claim has a corresponding measurable test and artifact.

Decision-gate criteria. Advancement from one aim phase to the next depends on pre-specified criteria: stability thresholds for transition indicators, minimum calibration standards in out-of-sample evaluation, and reproducibility checks for pipeline outputs. Criteria are intentionally conservative to prevent over-interpretation of early signals.

A major synthesis output from these aims is a phase diagram of wildfire behavior in reduced regime space, centered on transport-demand ratio (for example V/D) and connectivity, with extension axes for forcing anisotropy, dryness or effective resistance, coupling strength, and heterogeneity scale ratio where data support them. The phase diagram will classify fragmented, transitional, and coherent-continuous regimes with uncertainty bounds and will be released as both a figure-level and concept-level product.

3. Data, Measurements, and Preprocessing

Target length: 1,100 words. The empirical backbone of the project is event-based reconstruction and multivariate coupling. `FIRED` provides coherent wildfire events from satellite burned-area pixels, including temporal progression and event-level perimeters. `CubeDynamics` provides the computational grammar for organizing and transforming spatiotemporal arrays, enabling reproducible integration of perimeter trajectories with meteorology, fuels, and landscape covariates.

Our measurement system follows the repository's methods hierarchy. First, we derive canonical trajectory variables: $P(t)$, $A(t)$, dA/dt, and associated geometry descriptors. Second, we estimate scale-conditioned metrics such as perimeter-area relationships, roughness proxies, and anisotropy indicators. Third, we attach environmental context along the evolving interface (e.g., wind, stability, moisture proxies) to test coupled hypotheses instead of purely geometric correlations.

To unify observations and simulations, we represent growth using an ignition-time field $T(x, y)$, or equivalent arrival-time representation, defined on harmonized spatial grids. This common representation jointly encodes front position, spread progression, and geometry through time, enabling direct computation of diagnostics and direct comparison between observed and simulated fires without ad hoc output translation.

Data quality and uncertainty are treated as first-class modeling inputs. Every transformation step will preserve provenance metadata, quality flags, and versioned configuration. Missingness is addressed through sensitivity designs comparing complete-case analysis, principled imputation, and uncertainty-weighted estimators. Sensor and resolution dependence are quantified through perturbation analyses, allowing us to separate robust signatures from measurement artifacts.

The preprocessing architecture is intentionally modular. Components for perimeter extraction, geometry estimation, environmental coupling, and metric calculation are containerized as reproducible pipeline stages with documented interfaces. This allows partial progress even when one data stream is delayed and supports external reuse by collaborators who may only require subset functionality.

Deliverables from this component include Dataset v1 and v2 artifacts consistent with repository planning language: event-level time series, derived observables, uncertainty envelopes, and reproducible pipeline code with benchmark-ready outputs.

We will harmonize temporal alignment across data sources using explicit event clocks and resampling windows. This is necessary because sensor products, meteorological reanalysis, and fuel datasets operate at different temporal granularity and reporting conventions. Harmonization procedures will be documented and sensitivity-tested so that modeling outcomes are not artifacts of arbitrary alignment choices.

Measurement definitions will be explicitly versioned. For each key metric, we will track formula choices, smoothing assumptions, spatial discretization parameters, and treatment of disconnected geometry. This addresses a recurring comparability problem in wildfire analysis: two studies may report the same named metric while using materially different definitions.

Environmental coupling will be handled through interface-conditioned sampling rather than only event-level averages. This preserves directional and local context and supports tests of whether

particular boundary segments, rather than whole-fire means, are associated with transition behavior.

Quality control includes anomaly detection on trajectory continuity, geometric plausibility checks, and cross-source consistency tests. Events that fail minimum quality criteria will be flagged with transparent exclusion logic rather than silently dropped. We will maintain an audit trail of exclusions so downstream inference remains interpretable.

The final data products are designed for dual use: immediate project inference and external benchmark adoption. To support this, we will provide clear schemas, example notebooks, and machine-readable metadata that describe variable definitions, units, uncertainty fields, and processing lineage.

Data stewardship considerations are integrated from the start. For each source, we will track access constraints, usage permissions, and citation requirements, and we will maintain clear separation between openly shareable derivatives and restricted components. This allows transparent compliance while maximizing reproducibility where licensing permits.

Computational reproducibility is treated as a measurable output. Pipeline runs will capture configuration snapshots, dependency manifests, random seeds where applicable, and runtime provenance metadata. This makes it possible to reproduce benchmark outcomes and to audit differences when results diverge between environments.

4. Modeling Framework and Algorithmic Innovation

Target length: 1,300 words. The project evaluates wildfire dynamics through complementary model roles rather than a single monolithic simulator. The first role is local spread representation: process-informed models that capture ignition propagation, local coupling, and conditional heterogeneity effects. The second role is coherent-front representation: reduced geometric formulations that focus on boundary evolution once front continuity emerges. The third role is transition inference: statistical mechanisms that detect regime state and mediate handoff or weighted mixing between model roles.

This architecture is motivated by repository narrative synthesis and resubmission strategy. In pre-emergent and transition regimes, local spread assumptions are often indispensable; in post-emergent regimes, boundary geometry may explain a larger fraction of growth variability. The proposal therefore asks not "which model wins globally" but "which model class is sufficient under which regime conditions."

The modeling philosophy is explicitly theory-first generative modeling. We formulate minimal governing principles, instantiate them as candidate mechanisms, and test whether resulting growth on a structured substrate reproduces observed fire form. In this view, geometry and topology of the advancing interface are measurable records of process and therefore central inference targets.

As a scientific instrument for mechanism discrimination, we use a staged model hierarchy. M0 represents isotropic, fuel-controlled growth. M1 adds directional anisotropy from wind and slope. M2 adds coupling terms that organize and stabilize the advancing front. M3 adds extended dynamics where justified by evidence, including nonlocal effects and richer transport-demand coupling.

This hierarchy is not a convenience layer; it is the primary instrument for identifying which added ingredients are necessary in which regimes.

Across this hierarchy we seek reduction into a compact regime space rather than uncontrolled parameter growth. Candidate reduced controls include connectivity, dryness or effective resistance, forcing anisotropy, coupling strength, and heterogeneity scale ratio. Final dimensionless forms will be data-constrained, but the conceptual objective is explicit: reduce apparent complexity into a small number of testable controls.

Algorithmic innovation appears in four areas.

(i) Transition-aware gating and mixtures. We compare deterministic thresholds, probabilistic gating, and continuous-mixture formulations based on transition indicators. These alternatives are evaluated on both predictive metrics and interpretability of inferred state changes.

(ii) Mechanism-elimination experiments. Guided by the repository's generative framework, we stage model experiments that test necessary ingredients: dimensional adequacy, interface roughening, connectivity constraints, and geometry-throughput coupling. Models are retained only when they satisfy both exponent-level and geometry-level criteria within uncertainty.

(iii) Surrogate and emulator acceleration. We develop validated surrogates for expensive model components to support broad parameter sweeps and uncertainty decomposition. Surrogate use is constrained by strict domain-of-validity checks against reference runs.

(iv) Cross-model comparability. All model outputs are converted into common observable space so that comparisons occur on aligned measurements rather than model-internal variables. This reduces ambiguity and supports fair ranking.

The expected outcome is a constrained mechanism map: where local physics is necessary, where reduced geometry is sufficient, and where hybrid formulations provide the strongest explanatory and predictive tradeoffs.

Core figure outputs are named in advance to improve reviewability and execution clarity: the decisive single-law versus multi-regime comparison figure (Benchmark 0); a regime-classification time series with uncertainty; a transition-alignment figure comparing observed transition timing with V/D crossings; a phase diagram in reduced regime space; and a cross-model comparison panel stratified by regimes including transition and extreme-fire windows.

Model development will proceed in explicit tiers. Tier 1 focuses on robust baseline implementations and reproducible parameterization. Tier 2 introduces regime-dependent augmentation and transition logic. Tier 3 performs stress testing under extreme scenarios and cross-domain transfer checks. This staged design reduces integration risk and keeps evaluation synchronized with model complexity.

We will benchmark against established and commonly referenced model classes represented in repository documents, including operational and research simulators where compatible outputs are available, as well as statistical and null baselines. The goal is not to claim universal superiority, but to identify condition-specific advantages and failure modes under transparent protocols.

A central innovation is mechanism sufficiency mapping. Instead of ranking only by aggregate

score, we ask which mechanism families remain consistent with all validated constraints in each regime class. This creates a sharper scientific product: elimination of implausible explanations and bounded support for survivors.

We also formalize geometry-to-throughput coupling tests that link interface properties to growth rates. Competing formulations, such as dA/dt scaling with perimeter alone versus perimeter interacting with environmental state, will be compared with cross-validated and information-criterion approaches. This links geometric claims to dynamical consequences.

Extreme-event behavior receives dedicated attention. Many models can perform adequately in moderate conditions yet fail in directional, high-growth windows. We therefore include targeted extreme-regime tests using anisotropy, rapid-acceleration, and forcing alignment diagnostics. Mechanisms that fail these tests are not promoted as general explanations.

Finally, implementation choices prioritize portability. Model wrappers, configuration schemas, and output adapters will be designed so additional model families can be integrated with limited custom engineering. This supports long-term benchmark evolution beyond the grant window.

To prevent methodological drift, every model branch will be evaluated under harmonized input definitions and output extraction rules. Where models require distinct internal parameterizations, comparability will be preserved through standardized external observables and common scoring protocols.

The project also includes ablation-style experiments that selectively remove candidate mechanism components, such as connectivity constraints or environmental coupling terms, to test whether specific ingredients are necessary for reproducing observed structure. These tests help move from correlative fit toward causal plausibility.

5. Verification, Validation, and Uncertainty Quantification

Target length: 1,200 words. Verification and validation (V&V) are designed as the scientific core of this proposal, not an afterthought. Repository planning documents emphasize that success should be defined by reliable mechanism discrimination and transparent uncertainty, including in negative-result scenarios. Our V&V framework therefore combines predictive testing, structural diagnostics, and uncertainty calibration under shared protocols.

Predictive performance. We evaluate event-level and time-local performance under cross-validation and held-out scenario tests, with special emphasis on transition-heavy windows. Metrics include trajectory error, distributional divergence, and time-warp-aware alignment where appropriate.

Structural fidelity. Because aggregate error can mask mechanism failure, we require models to reproduce regime-relevant structure: continuity changes, roughness evolution, perimeter-area behavior, and anisotropy signatures. This is where benchmark language from repository model pages is directly operationalized.

Uncertainty calibration. We decompose uncertainty into observational, parametric, and model-form components. Calibration diagnostics evaluate whether uncertainty intervals are informative

and reliable, not merely wide. Reporting will distinguish uncertainty that may be reduced through better data/modeling from uncertainty likely intrinsic to regime variability.

Identifiability and robustness. Synthetic validation experiments (with known generative conditions) test whether the full pipeline can recover intended mechanisms and avoid false confidence. Global sensitivity analyses evaluate whether "successful" behavior requires fragile tuning or persists across meaningful parameter ranges.

Failure reporting. If mechanisms cannot be distinguished within uncertainty, or if findings do not generalize across datasets, those outcomes are reported as substantive results. This aligns with repository evaluation language and strengthens scientific integrity.

Verification tasks will include implementation consistency checks, unit and integration test coverage for key metric modules, and reproducibility checks across compute environments where feasible. Validation tasks then operate on scientific behavior, not merely code execution. This separation reduces silent technical bias in scientific conclusions.

For model-data comparison, we will use a portfolio of complementary metrics rather than a single score. Repository planning materials already reference measures such as RMSE, KS distance, DTW alignment, and Earth Mover's Distance. These metrics capture different error structures and reduce overfitting to one objective.

We will explicitly test sensitivity to data definition choices, including boundary extraction rules, temporal aggregation windows, and smoothing settings. If conclusions depend strongly on one preprocessing variant, claims will be downgraded until robustness is demonstrated.

Uncertainty communication outputs will include visual and tabular templates designed for both technical and semi-technical audiences. The purpose is not only statistical completeness but interpretation discipline: users should understand what is known, what is uncertain, and what decisions should remain conditional.

Generalization is evaluated across event classes, geographies, and forcing regimes where data coverage permits. Where generalization is weak, we will report applicability domains rather than broad claims.

Validation reporting will be organized in tiers for different audiences. Technical reports will include full metric distributions, sensitivity results, and calibration diagnostics. Synthesis reports will provide concise decision-oriented interpretations with clear caveats. This two-tier approach preserves rigor while improving usability.

We will also maintain a "known limitations" registry that is updated as evidence accumulates. This registry will document unresolved ambiguities, conditions with weak support, and model classes that fail specific criteria. Maintaining this registry as a first-class product supports scientific transparency and helps external users avoid overgeneralization.

Proposal-level acceptance criteria are stated directly. Evidence should favor multi-regime over single-law explanations in a majority of tested fires within analyzed domains; inferred regime transitions should align with V/D transitions within a predefined temporal tolerance; reduced geometric models should reproduce post-emergent behavior within stated uncertainty; and the benchmark suite should differentiate models on transition and extreme-fire behavior, not only aggregate fit.

If these criteria are not met, the negative result remains a substantive contribution: it constrains plausible mechanism space, bounds proxy usefulness, and clarifies limits of current wildfire modeling frameworks.

6. Work Plan, Milestones, and Deliverables

Target length: 1,000 words. The work plan follows the Detect–Explain–Apply sequencing already established in repository research-program materials and is implemented as decision-gated annual phases.

Year 1 (Detect). Primary objectives are data system assembly, baseline metric production, and transition indicator validation. Milestones include Dataset v1 with core observables ($P(t)$, $A(t)$, dA/dt, geometry, anisotropy), an initial comparison engine, and a first model-selection report contrasting single-regime versus multi-regime framing.

Year 2 (Explain). Primary objectives are mechanism experiments, hybrid model implementation, and uncertainty-aware comparison across model classes. Milestones include Model Library v1, Validation Framework v1, and evidence-weighted regime-specific sufficiency maps.

Year 3 (Apply). Primary objectives are benchmark formalization, open release packaging, and translational synthesis for external users. Milestones include final benchmark artifacts, reproducibility bundles, and a consolidated mechanism synthesis report.

Deliverables are tracked as two coordinated streams. Scientific deliverables include transition tests, mechanism discrimination outcomes, and regime-specific inference results. Infrastructure deliverables include versioned datasets, pipeline code, benchmark utilities, and documentation packages. This dual-stream design ensures durable value even under partial or mixed scientific outcomes.

Explicit artifacts include a fire growth trajectory dataset derived from satellite observations; versioned regime labels with uncertainty; V/D and related transport-demand proxy datasets; CubeDynamics-integrated scaling diagnostics; a model experiment library covering M0–M3 and ablation variants; a benchmark suite for existing and future fire models; and a consolidated theoretical framework for wildfire growth geometry anchored in measurable evidence.

Management cadence mirrors repository staffing and evaluation plans: weekly discrepancy reviews across measurement-model-validation teams, quarterly milestone assessments, and explicit risk tracking with contingency triggers for data bottlenecks, computational delays, and underperforming model branches.

Within each year, we define quarter-scale decision points. Early quarters emphasize pipeline stability and measurement reliability. Mid-year quarters emphasize comparative analysis and model discrimination. End-of-year quarters emphasize synthesis, release readiness, and conference manuscript alignment. This quarterly structure prevents milestone drift and supports corrective action before annual reporting windows.

Benchmark 0 is an explicit work-plan gate rather than background context. It is executed as a formal deliverable with AIC/BIC comparison, cross-validation, change-point analysis, and alignment tests against transport-demand transitions. Broad claims about regime structure are contingent on

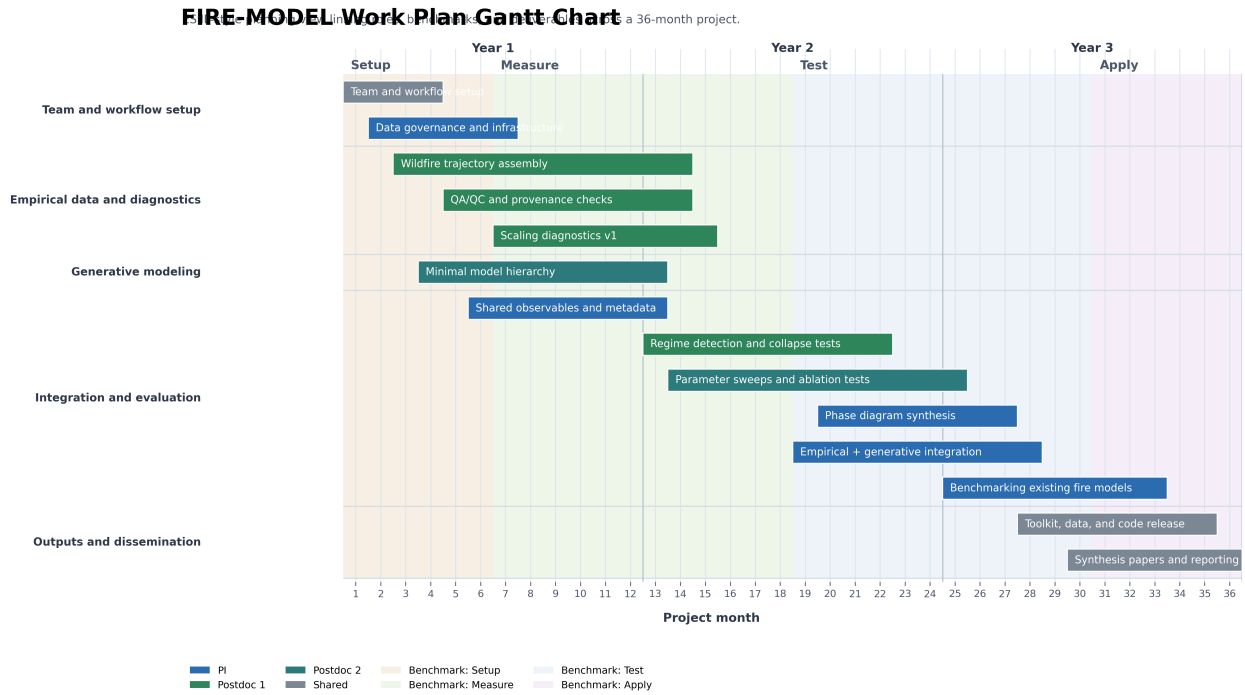

Figure 1: FIRE-MODEL three-year Gantt timeline generated from the repository planning workflow.

passing this gate.

Conference and manuscript outputs are tied to milestone maturity, not arbitrary dates. For example, early measurement papers depend on Dataset v1 stability; model-comparison papers depend on completion of cross-model evaluation and uncertainty diagnostics. This improves coherence between narrative outputs and demonstrated evidence.

Risk governance includes explicit fallback plans. If hiring timelines shift, core pipeline and benchmark foundations will be advanced by PI and collaborator effort using modular task decomposition. If data coverage is uneven, we prioritize robust domain-specific claims over weak generalization.

We will maintain a live milestone dashboard in repository documentation to track status, blockers, and evidence artifacts for each deliverable. This supports internal accountability and efficient resubmission updates.

Deliverable packaging includes explicit handoff standards: each major artifact must include a plain-language summary, technical specification, reproducibility instructions, and validation status. This reduces integration friction across workstreams and improves external reuse.

At project close, we will deliver a consolidated evidence book that links hypotheses, diagnostics, model tests, uncertainty findings, and milestone artifacts. This product is intended to make reviewer and stakeholder evaluation efficient and transparent.

7. Team Integration, Convergence, and Management

Target length: 700 words. The team structure follows the repository staffing model: two complementary postdoctoral tracks and PI-level integration. One track focuses on computational geospatial measurement and event-level data quality. The second focuses on complex systems modeling and mechanism experiments. The PI and senior collaborators coordinate convergence logic, milestone governance, and synthesis.

Convergence is defined operationally, not rhetorically. No claim advances without aligned evidence across three artifacts: measured observables, model outputs in common metric space, and uncertainty diagnostics. This artifact-centered review process reduces disciplinary drift, enforces comparability, and creates auditable reasoning chains from data to conclusion.

Role-specific outputs map to project milestones. Measurement leadership delivers dataset versions, geometry diagnostics, and uncertainty envelopes. Modeling leadership delivers mechanism libraries, comparative experiments, and transition-aware formulations. Integration leadership delivers cross-model evaluation, risk resolution, and proposal-level synthesis products. Annual outputs include conference dissemination and first-author manuscript progression, as specified in planning documents.

Risk mitigation is built into modular dependencies: measurement, modeling, and validation can proceed in parallel with iterative synchronization. If one stream slows, other streams generate actionable diagnostics that guide adaptation rather than causing full stoppage.

Team operation includes weekly technical integration meetings, monthly synthesis meetings, and quarterly milestone reviews. Weekly meetings resolve discrepancies between measurement and modeling outputs. Monthly meetings evaluate scientific implications and adjust experiment priorities. Quarterly reviews confirm milestone completion against predefined acceptance criteria.

Mentoring and professional development are explicit project responsibilities. Postdoctoral researchers will lead major deliverables, manage reproducible workflows, and present at conferences as first authors where appropriate. This approach builds workforce capacity while strengthening project execution.

External advisory input will be used selectively for methods and interpretation checkpoints, especially around uncertainty communication and benchmark usability. Feedback will be documented and translated into explicit action items.

Coordination protocols include explicit interfaces between personnel roles. Measurement teams provide versioned data products on scheduled release cycles; modeling teams provide benchmark-ready outputs with standardized metadata; validation teams provide comparative scorecards and uncertainty summaries. These interfaces reduce coordination ambiguity and make dependencies testable.

The management structure also emphasizes continuity planning. Key workflows will be documented at a level that allows role transitions without loss of institutional memory, supporting resilience under staffing changes and improving long-term project sustainability.

8. Broader Impacts Plan

Target length: 700 words. Broader impacts are structured as an integrated system rather than isolated outreach activities. Repository planning materials identify three coordinated components: an interactive wildfire exploration platform, narrative media and educational products, and an open science training ecosystem.

Interactive component. The Fire Lab concept provides a user-facing interface for exploring model behavior, uncertainty, and scenario structure. Its purpose is not operational command support, but interpretable scientific engagement for students, researchers, and agency technical audiences.

Narrative and education component. Partnership with Impact Media Lab supports production of documentary-style and instructional materials that translate technical wildfire modeling ideas into accessible narratives while preserving scientific fidelity. Companion lesson materials connect conceptual framing, visual outputs, and mechanism-testing logic.

Open science component. Code, workflows, and benchmark assets are released with documentation and training modules. Workshops combine conceptual orientation, hands-on pipeline use, and feedback capture, creating bidirectional learning between developers and users.

Evaluation will use concrete indicators: workshop participation and outcomes, tool adoption, repository usage metrics, external method reuse, and evidence of educational uptake. The broader-impact hypothesis is that transparent, reusable mechanism-based tools can improve both scientific literacy and practical interpretation of wildfire modeling outputs.

Broader impacts are designed for continuity across audience types. Scientific users access reproducible methods and benchmark resources. Technical agency users access interpretable diagnostics and uncertainty templates. Educational users access structured narrative and instructional material. These layers are linked so that one set of outputs supports multiple pathways.

User pathways are defined explicitly. Direct users are fire model developers, remote sensing scientists, fire behavior researchers, and Earth system scientists. Secondary users are agency research groups and strategic planning or risk-assessment teams. The operational pathway is intentionally described as indirect: scientific insight $\rightarrow$ improved models $\rightarrow$ better forecasting tools $\rightarrow$ downstream operational benefit.

Workshops serve both dissemination and co-design functions. Participants will test tools, report friction points, and suggest scenario classes that matter for practice. This feedback will inform subsequent platform and documentation updates, creating an iterative improvement loop.

We will prioritize accessibility in communication artifacts through clear visual conventions, plain-language summaries, and modular learning materials that can be adapted for different educational contexts. These choices are intended to broaden participation in wildfire modeling literacy.

Long-term impact is defined as durable infrastructure and reusable practice: tools that continue to be used, workflows that are adopted, and conceptual frameworks that shape subsequent wildfire model development.

To support durability, we will align dissemination formats with likely user pathways: repository

releases for technical users, structured curricula for educators, and concise narrative products for broader public audiences. These formats draw from the same underlying evidence base, reducing contradiction between technical and public-facing outputs.

We will assess sustainability through post-release indicators, including updates by external contributors, replication of workshop exercises, and reuse of benchmark templates in independent studies. These indicators provide a practical test of whether broader impacts persist beyond initial launch.

9. Results from Prior NSF Support and Positioning

Target length: 500 words. This section will be finalized with award-specific factual language at submission time. The draft framing is as follows: prior work established key enabling infrastructure, including event-level fire reconstruction workflows, scalable data-cube analytics, and methodological prototypes for geometry-aware evaluation. Those achievements revealed both opportunity and limitation: rich data and computational tools exist, but mechanism discrimination under regime change remains underdeveloped.

The present proposal is positioned as the next logical step: move from descriptive and component-level advances to a falsifiable, integrated inferential system that can test regime hypotheses, compare model classes under aligned diagnostics, and report uncertainty with decision-relevant clarity.

Positioning strength comes from the repository's existing integration: funder-aligned requirement tracking, detailed resubmission strategy, method-level technical notes, staffing and milestone plans, and reproducibility-oriented workflow documentation. Together these assets demonstrate readiness to execute a rigorous FIRE-MODEL project with strong potential for field-level adoption.

In final submission form, this section will include concise evidence chains from prior outcomes to present feasibility claims: completed infrastructure, demonstrated workflow reliability, publication and dissemination outcomes, and lessons that directly informed this project's design decisions. The purpose is to show continuity of capability while making clear that the proposed research question is substantively new.

This section will also transparently identify remaining capability gaps and how the proposed work addresses them. Doing so strengthens credibility by avoiding overstatement and framing the project as a targeted response to known limitations rather than a broad aspirational effort without operational grounding.

In the final proposal package, this section will include concrete placeholders for award numbers, principal outputs, trainee outcomes, and prior dissemination results. Those entries will be limited to verifiable facts and aligned with sponsor formatting rules. The narrative purpose will remain consistent: demonstrate readiness through evidence, delimit claims to supported scope, and show why this next-step project is both ambitious and feasible.

Required Companion Documents (Outside this narrative)

- References Cited

- Data Management and Sharing Plan
- Postdoctoral Mentoring Plan (if applicable)
- Biographical Sketches (SciENcv)
- Current and Pending Support (SciENcv)
- Facilities, Equipment, and Other Resources

Word-Count Target Summary

- Project Summary: Overview — 300
- Project Summary: Intellectual Merit — 225
- Project Summary: Broader Impacts — 225
- Introduction/Motivation — 800
- Hypothesis + Aims — 900
- Data + Measurements — 1,100
- Modeling Framework — 1,300
- V&V + UQ — 1,200
- Work Plan + Deliverables — 1,000
- Team + Convergence — 700
- Broader Impacts Plan — 700
- Prior NSF + Positioning — 500
- **Total target across sections — 8,950**